

The Design of COVID Alert NY in Six Parables

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ABSTRACT

A multi-disciplinary team of volunteers collaborated with New York State to design the COVID contact tracing app for its residents (*COVID Alert NY*). Facing a myriad of constraints and conflicting forces, we conducted 13 evaluation studies, using 5 UX methods, in 10 weeks. These studies revealed over 150 usability and attitudinal issues, many of which were prioritized and addressed before launch on Oct 1, 2020. We selected six stories in which to convey some of the lessons we learned: parables, if you will, for UX practice.

CCS CONCEPTS

• Human-centered computing; • Interaction design; • Interaction design process and methods;

KEYWORDS

COVID-19, research methods, mobile app design

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1 INTRODUCTION

In mid-July 2020, a group of volunteers with Tech:NYC teamed with New York State (NYS) to develop a digital contact tracing app for smartphones.¹ The primary purpose of the app is to alert users when they have been in close contact² with another app user who has tested positive for COVID-19, so they can quarantine and break the chain of exposure. This app is part of a broader contact tracing program in NYS that includes thousands of human contact tracers and the technology and protocol for communicating with New Yorkers.

NYS had contracted with Nearform Ltd, the company that developed the Irish contact tracing app (*COVID Tracker*), which launched in early July 2020. Thus, *COVID Tracker* was our team's starting point for UX design and development. Nearform used a decentralized approach with the Apple/Google exposure notification technology. The UX design problem was to modify *COVID Tracker* for

¹The app, *COVID Alert NY*, can be downloaded from the Apple App Store and Google Play <https://coronavirus.health.ny.gov/covid-alert-ny>.

²In NYS, the definition of "close contact" is being within 6 feet of a person for 10 minutes or more.

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the NYS populace in parallel with the engineering work required to integrate it into the NYS system.

Table 1 lists the UX methods we performed in the 10 weeks between when we joined the project and launch of the app; Figure 1 shows the timeline. We did 13 studies using 5 different methods revealing over 150 unique usability issues. We rapidly prioritized and iterated on the design throughout.

The next section briefly lays out the characteristics of the project and the principle that drove decisions about which UX processes to pursue and the changes to design and implement. Then we tell six parables of specific research results and design responses that taught us lessons about those decisions and our processes. We conclude with a summary of general lessons learned.

2 OUR GUIDING PRINCIPLE AND PROJECT CHARACTERISTICS

The principle guiding all our work was "*UX never holds up the release of the app.*"

Every day that release was delayed could mean lives and livelihoods lost. Although UX details turn out to matter in the adoption of contact tracing apps [2], the *Covid Tracker* was already working in Ireland, so why not just release it in NYS as is? The project required other time-consuming tasks, such as integration with the NYS Department of Health databases, software security checks, the creation of advertising, and political agreements with neighboring states. UX would work in parallel with these activities to make as much progress as possible before release.

As the release schedule changed because of these other tasks, and evaluation opportunities arose, we decided which evaluation method to use by weighing its estimated time to complete vs the value of the data it could provide. As UX data came in, we prioritized the most important issues to tackle. We were prepared to freeze the design at any point, confident that we had done the best for the users in the time available.

Three characteristics of this project worked for and against this principle:

A pre-defined starting point. Before embarking on UX research, Nearform's *COVID Tracker* app from Ireland was chosen as the underlying technology for *COVID Tracker NY*. While we were free to suggest modifications to the system, having a functional baseline exerted a strong influence on process and design decisions throughout the project.

Urgency. From the moment we started the project, there was a climate of extreme urgency. The first release was always imminent: a "constant 2 weeks away." We could not plan a series of UX methods or design responses that we knew would take weeks to accomplish. We had to make decisions based on only having a week or two to bring value to the users.

One implication of this urgency was that we had to forgo the preparatory literature search and reading that precedes a project in a

Table 1: UX methods used in the *COVID Alert NY* project

UX Method	Design Version	N	Participant Source and Demographics	No. of Issues
Cognitive Walkthrough [1]	<i>COVID Tracker</i>	n/a	n/a	23
Paper-based moderated think-aloud	proto.v1	7	Convenience (5 >60yr: 4 female: 2 Latinx)	29
Paper-based moderated think-aloud	proto.v2	4	Convenience (1 >60yr: 4 female)	15
Wizard of Oz moderated think-aloud	proto.v2	6	Convenience (4 >60yr: 3 female)	38
Remote unmoderated think-aloud	proto.v3	10	UserZoom Panel (0 >60yr: 5 female: 2 Asian, 4 Latinx)	8
Remote unmoderated think-aloud	proto.v4	10	UserZoom Panel (0 >60yr: 8 female: 1 Native American, 1 Asian, 3 Latinx, 4 Black, 1 other))	21
Remote unmoderated think-aloud	proto.v5	10	UserZoom Panel (0 >60yr: 7 female: 2 Asian, 1 Latinx, 1 Black)	12
Remote unmoderated think-aloud	proto.v6	10	UserZoom Panel (10 >60yr: 3 female: race not asked)	1
Remote unmoderated think-aloud	proto.v7	10	UserZoom Panel (10 >60yr: 5 female: race not asked)	n/a
Remote unmoderated think-aloud	proto.v8	10	UserZoom Panel (10 >60yr: 6 female: race not asked)	1
Remote unmoderated think-aloud	proto.v9	8	UserZoom Panel (4 >60yr: 7 female: race not asked)	n/a
Columbia University field trial ³	0.0.20-0.0.37	151	Faculty and graduate students	33
SUNY field trial	0.0.38-0.42	769	Primarily undergraduate students	49
Reading Feedback & Reviews	1.0.0	n/a	Self-selected & unknowable	

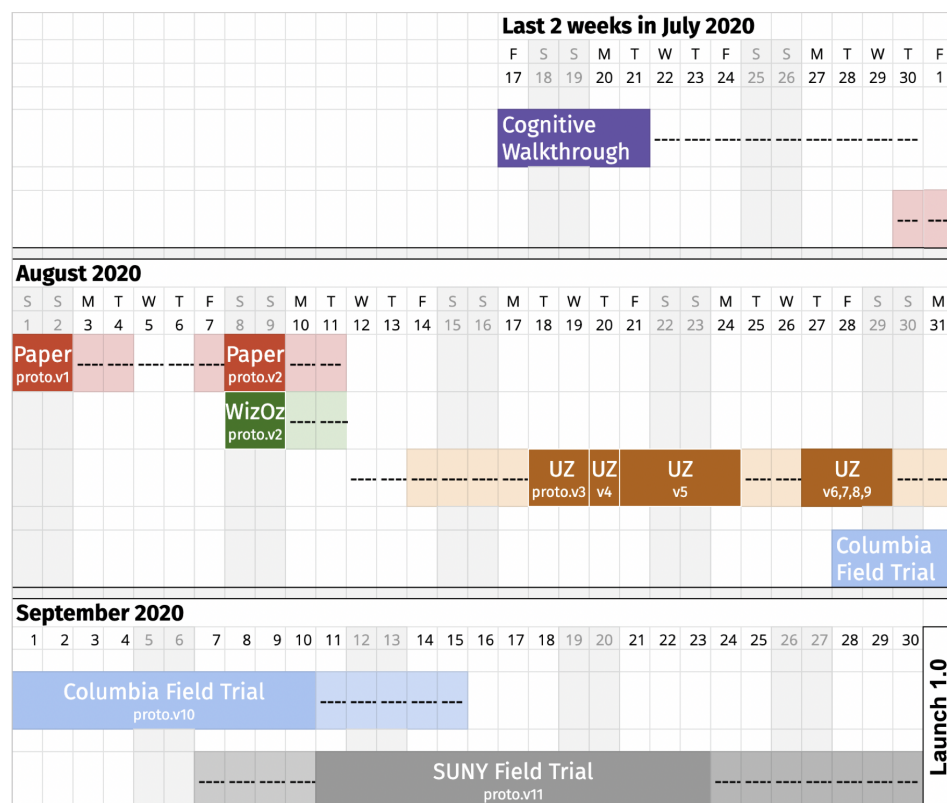


Figure 1: Rectangles of saturated color indicate the analysis of Cognitive Walkthrough or the data collection phases of the empirical methods. The less saturated color indicates times when we were preparing or analyzing. The dashed lines mean we were designing and implementing a new prototype.

new area. Instead, we needed “just in time expertise.” Team members drew on their networks to get help from experts in subfields like usable privacy and security, survey design, and statistics, asking them to volunteer an hour or two to help us shape our processes and designs. We thank these generous friends profusely.

Trusted Team. The core team, defined as the folks who met every day in the standup, were a small but diverse group. “T-shaped” professionals, all contributed to design discussions, with domain expertise represented by one mobile implementer, one mobile designer, one UX researcher expert in formative analytic and empirical usability methods, one documentation writer, one service designer, and one interaction designer who served as the project manager. Not everyone knew each other from previous work, but every member had worked with at least one other member sufficiently to be trusted. Trust was symmetric and transitive. This team started performing immediately with only short and intermittent bouts of forming, storming and norming [3].

We learned a lot working under this principle and these constraints, and have selected the following six stories to convey some of the most important lessons learned.

3 PARABLES FOR UX PRACTICE

3.1 Analysis First: the Story of Cognitive Walkthrough

When we started evaluation and redesign, we believed that the NYS app would be released in two or three weeks, which meant we had only about a week for evaluation, redesign, and implementation. Although we had *COVID Tracker* to start from, this timeline seemed too short to assemble, conduct, and analyze an empirical usability study, especially amid the restrictions of a pandemic.

Thus, we turned first to an analytic method, Cognitive Walkthrough (CW) [1], to assess the usability of *COVID Tracker*. Since CW was developed to rapidly assess walk-up-and-use systems, it seemed perfect for our needs. Over our first weekend, the author and two team members took screenshots of the *COVID Tracker* as they installed and enabled the app, explored the COVID-19 data it provided, and worked through features such as its symptom recording capability. For features we could not access ourselves (i.e., a close contact alert and responding to a positive test for COVID-19), we took screenshots from a YouTube video by the Irish HSE [4]. For each action available in the app, we answered CW’s four questions, assessing the match between the user’s goal and the flow of the system, the perceptibility of the available actions, the match between labels and the user’s goal, and the feedback provided after each action.

We modified the traditional CW procedure by analyzing every path available to the user instead of just the correct path, because the inability to recover from detours or errors would be as detrimental to a user’s health and safety as a difficult “happy path.” We also added a question: Does this copy and/or button label use appropriate language (e.g., no medical or technical jargon nor prose above a 6th grade reading level)?

We got many benefits from CW. By systematically examining every screen and action, we all gained a deep understanding of the app we were starting from. Reviewing the CW questions together created a shared understanding and revealed the different perspectives we each brought to the project. As expected, the CW found

usability issues at many levels (23 in all), from detailed visual design to mismatches between users’ goals and the flow of the app. For example, the CW showed us the importance of “compact design” when important information was not visible on smaller phones. Our additional fifth question about appropriate language highlighted the copy and button labels that needed to change and encouraged us to evaluate all text we proposed during redesign.

One result was that *COVID Tracker*’s onboarding process required eight actions and had screens that did not align with the user’s goal of setting up the app. Three of the extraneous screens talked about how the app worked and data privacy. We were able to slim the onboarding process down to five steps—with 2/3rds of the text of the original—by calling an expert in usable security and privacy, who graciously explained design guidelines that helped us provide just-in-time information to our users.

Perhaps the most profound change in onboarding was inspired by the very first CW question on the very first screen users see. On that screen, *COVID Tracker* asks users if they are 16 or older⁴. If a user says “no,” they are told they cannot use the app and are barred from proceeding. This is a goal mismatch between a minor’s desire to help stop the spread of COVID-19 (inferred from them having downloaded the app) and the system’s kicking them out. Because we examined this “incorrect” action, we were inspired to think more broadly about goals of using the app. With schools re-opening across NYS, many parents want their children to be alerted to an exposure and might therefore want this app on their children’s phones. This “age gate” was forcing the parents or the children to lie in order to proceed. We presented the issue to NYS with the result that *COVID Alert NY* now asks the user to confirm “I am over 18, or I am the parent or guardian of a minor confirming that they can use this app.” (See Figure 2) Although it adds more text to read, we believe the rewrite not only allows civic-minded minors to use the app, but also may remind adults to think about protecting us all by putting the app on their children’s phones.

Thus, CW, an analysis based on sound psychology, backed by decades of use, was a great choice to begin this project.

⁴In Ireland, a person 16 or older can consent to medical treatment. [5]

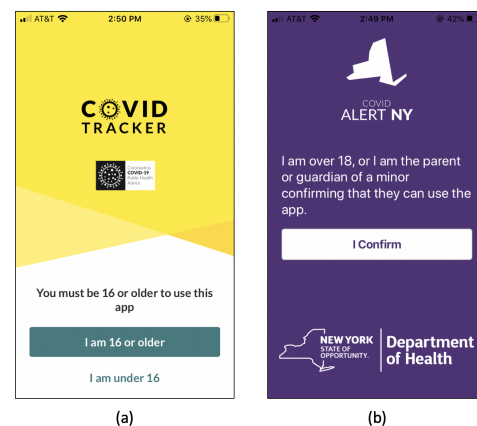


Figure 2: “Age gate” for *COVID Tracker* (a) and *COVID Alert NY* (b), captured on October 10, 2020, Apache License v 2.0.

3.2 Making Do Can Do a Lot: The Story of Our Paper-Based Studies

After the CW, the team prepared, discussed, iterated, and documented design responses in an InVision prototype. In parallel, the author explored how to do think-aloud usability testing on the designs in the midst of a pandemic, just in case the release date slipped and we had time for empirical testing. Indeed, the release date was pushed out for reasons not involving UX, and guerrilla usability testing had a chance to contribute to the design.

The fastest formative empirical method is often paper prototyping, but using this face-to-face technique during a pandemic posed obstacles. The author devised a protocol where she printed out each question, task, and screen from the prototype. She fit them to the full height of 8.5x11 paper, which she confirmed to be readable from six feet away, even for people over 70 years old. Conducting the sessions outdoors, her equipment included a tape measure to mark six-foot spacing on a park bench, a mask and face shield, several rocks to weigh down the papers in case of wind, and a lovely canine assistant to make sure the participants stayed socially distant (Figure 3). Since no other researchers lived within walking distance to help with notetaking, she used a visual data collection sheet [6] (Figure 4) that allowed her to record the participants' answers to questions, actions, and comments while manipulating the paper screenshots.

The author conducted most of these sessions with a convenience sample of people who walk their dogs in Central Park at the same time each morning. Other sessions came from asking people who were enjoying the summer weather in the park if they would donate 15 minutes of their time to help evaluate the app (about one in five people agreed).

As expected, these sessions provided a wealth of information. The tasks were overwhelmingly seen as easy (SEQ [7] average of 6.5 for onboarding and 6.3 for responding to a COVID alert). Of course, there were findings that some text was confusing and some actions could not be found; these fed into the next design iteration.

We collected reactions to the data presented in *COVID Tracker*. Participants overwhelmingly preferred to see trends over time (7

of 11 participants), so *COVID Alert NY* shows data in graphs rather than absolute numbers. The sessions also revealed concerns the participants had about installing this app on their personal phones when it became available. For example, a common concern was that the app tracks a person's location (which it does not), a misconception that plagued us through all our studies and into the launch. We would translate these concerns into survey questions in subsequent studies to see if larger and more diverse populations shared these concerns.

The most surprising result came from the accidental demographics of this convenience sample. Most of the participants were over 60, as that is the age range of the author's dog-walking friends. In a poignant reminder that "the user is not like me," evidence emerged that common mobile design patterns cause problems for people over 60. For example, several of the older participants did not know that dots at the bottom of a screen invite you to swipe. Others did not know to look for a "<" at the top left to go back to the previous screen or that an "X" would close an overlay. One participant did not know to scroll to see more contents on a clipped screen. Some of these problems spurred immediate redesign, e.g., putting "Next" and "Previous" buttons beside the dots at the bottom of a screen. Others were deprioritized because adhering to mobile design conventions has advantages for those who know them and, if we had time to run more studies, more data might have emerged to refute this surprising result among this vulnerable population.

3.3 A Bridge Too Far: The Story of Hypothesis Testing With Older Users

At the end of the two weekends of guerrilla usability testing, the release date was again pushed out for reasons not involving UX. Luckily, we had been given access to a UserZoom account allowing us to do unmoderated remote think-aloud usability studies with the prototype presented on the participant's phone. We quickly learned enough about the features of UserZoom to conduct three 10-participant studies in a week. We could design and pilot a study in a day, then launch it to UserZoom's panel and have dozens of session videos waiting for us to analyze the next morning. UserZoom's



Figure 3: The author conducting a paper-based usability test.

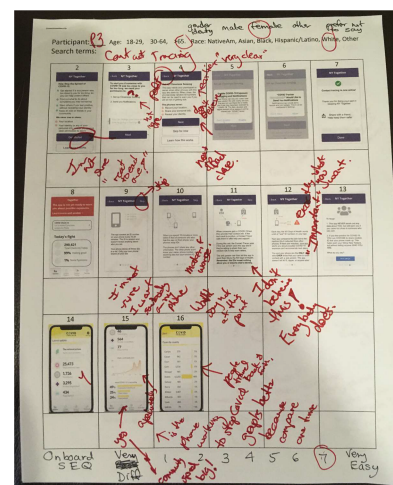


Figure 4: A visual data collection sheet.

panel allowed us to reach participants outside New York City, of different age groups, different ethnicities, and different socio-economic backgrounds from the convenience samples of our guerrilla studies.

The most critical new finding from analyzing these videos was that when participants got a COVID alert, they could get to the screen that told them to stay home, but they never tapped on the “Learn how to keep others safe” button. Therefore, they never saw detailed information about how to quarantine and did not see that NYS could provide support. In response, we put that information directly on the screen that resulted from tapping the notification, removing the need for an intermediate button tap.

Given the high-quality think-aloud data we were getting, we tried to take remote testing one step further to hypothesis testing. The guerrilla studies had indicated that older adults may have trouble with mobile design patterns, but that might have been an artifact of the paper-based and ‘Wizard of Oz’ techniques we used. We wanted more data, presented on the participant’s own phone, before making sweeping changes to the app. We designed three different versions of the app with varying degrees of mobile app patterns versus explicit buttons and labels. These sessions targeted participants over 60 years old. We ran 38 participants in this between-subject study in three days.

Unfortunately, we could analyze only 18 of the 38 videos before designs had to be finalized. These data convinced us that, at least for elders on the UserZoom panel, scrolling was not the problem we had seen in previous test formats, so we could move forward with designs that required scrolling on the smallest phone screens. Labeled buttons were easily used by all participants. However, the use of the <Back button, a problem for our elder participants in the guerrilla tests, also caused problems for these participants. A version that had both a <Back button in the upper left and a labeled button in the body of the screen performed better, but not statistically significantly better. Also, this design would require using a non-standard mobile pattern which might have adverse effects on the perception of the app among users in other age groups. The team could not reach consensus without additional performance and attitudinal data, which we just did not have time to collect and analyze. Therefore, we decided to stick with standard mobile design patterns.

3.4 Small Fixes Can Have Big Impact: The Story of Field Trials and the SMS Copy Change

Twenty-six percent of the participants in the Columbia field trial said it was difficult to enter the 6-digit number provided in an SMS when they received a simulated positive test for COVID-19. It is very important to enter this number because that is what allows your close contacts to receive alerts. Although in real life, most

people would be on the phone with a public health representative who could walk them through the number-entry procedure, some users will need to enter it without help and 26% was just too high to ignore. In free-text feedback, most of the participants reported that it was hard to find where in the app to enter the numbers and some commented that the SMS text was unclear.

It would be best if the user could just tap the SMS to enter the code (as suggested by some participants), but that would take a lot of engineering time that we just didn’t have before release. So we looked at the prose of the SMS to see if there was something we could do quickly to help. (Figure 5(a).) As reported by participants, the prose wasn’t explicit about all the steps the user had to take to enter the 6-digit number. We re-wrote the prose to include more of the steps and cues, including words like “tap” to direct their attention to the tabs at the bottom of the screen, “tap” to indicate a button, and explicitly telling them to scroll. (Figure 5(b).) We had to stay within a 160-character limit to comply with mobile design guidelines, so we couldn’t fit in the very last step of tapping the button labeled “Share Your Close Contact Codes.” We hoped that getting a user to scroll to the bottom of the screen, where a dark purple call-to-action button was, might get them closed enough.

This change was implemented for the field study at SUNY. This time only 5% of the participants said entering the number was difficult. The difference is significant ($p < 0.001$). Apropos of our design compromise, only 1 person (of 769) mentioned that the SMS was incomplete, saying “It did not say you must click a button to see the enter box for the code.”

This result suggests that this simple design change had a huge effect. However, maybe the difference was because the populations were different, e.g., maybe SUNY undergrads find everything easier than Columbia grad students do. To test this, we looked at the ease-of-use survey question for a task that we didn’t change between the two studies: Checking the COVID Alert. This time the difference between the two groups (16% difficult for Columbia, 10% for SUNY) was not significant ($p = 0.37$). This suggests that, for perceptions of the usability for the same app feature, the two populations are not different. Therefore, we have no evidence not to trust the result about the SMS design change.

The design principle we took from this is to make the instructions as complete as possible within mobile guidelines. Hence, when the communications arm of the project decided to start the message with “COVID Alert NY” instead of “NYS Department of Health,” we had 10 extra characters and could make the instructions even more complete, getting the users all the way to the screen that has the digit-entry box (Figure 5(c)).

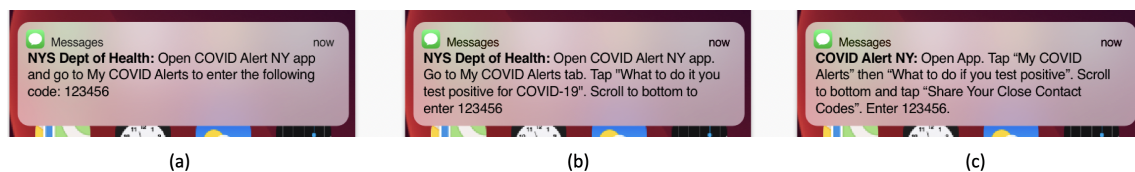


Figure 5: Three versions of SMS copy providing instructions on how to enter a 6-digit number into the app.

3.5 “Lean Startup” Approach to Testing: Build-Measure-Learn in Test Design Itself

When we started this project, the team members had substantial experience in face-to-face think-aloud usability testing but no experience in remote unmoderated testing or field trials. Thus, as the sliding release data afforded the opportunity to conduct such studies, we shifted into high gear to learn what we needed to know to get valuable data quickly. We practiced what might be considered a “lean startup” approach [8] to the design of our tests, where “build, measure, learn” helped us improve our study design and data collection with each iteration.

Starting with the guerilla think-alouds, our experience with every study purposefully contributed to the design of the next. Both the content and processes of early studies fed into the design of subsequent studies. For example, early think-alouds uncovered concerns that went through participants’ heads as they did the tasks, among them, fear of getting too many notifications because of “those joggers running past huffing and puffing without masks” (*the app only alerts if contact is within 6 feet for 10 minutes or more*), the app sharing personal data without their permission (*it doesn’t*), and the app tracking their location (*it doesn’t*). Concerns from the guerrilla studies were turned into check-all-that-apply questions in the remote unmoderated studies and exit surveys in the field trials.

We started the remote unmoderated studies with a “Basic” study on UserZoom (UZ), which had limited features for defining tasks and asking questions. This allowed us to (1) learn the basics of remote unmoderated technology, (2) verify that we could get high-quality think-aloud data in an unmoderated study equivalent to the moderated guerilla studies and (3) reach a broader range of demographics. We experimented with NPS [9] and SEQ [7] questions and found their numbers compared well to the participants’ behavior and comments, so we kept using them in subsequent studies. Our UZ studies became more sophisticated as we moved to an Advanced UZ contract and we incorporated more complex logic. Of course, we pilot tested each study with “friends and family,” often recruiting them from three times zones away as we raced to ready a study for launch the next morning. After each study, we examined the freeform comments associated with survey questions to find evidence that questions were being interpreted differently by the participants than we had intended; we rewrote those in response to the data.

We also experimented with recruiting segments and screening criteria, finding the fastest way to fill the needs we had. For example, for finding participants outside New York City, we first recruited from NY State only, but got too few participants in a day, so, since the app interoperates with surrounding states (New Jersey, Pennsylvania, and Delaware) we recruited from those states as well to fulfill our need. In addition, we had been told by UZ personnel that participants rarely do studies on weekends, and we found this to be especially true of older panelists. We had to keep our study open a full day longer than we had hoped in order to get sufficient participation.

Thus, in addition to collecting valuable data for the development of *COVID Alert NY*, we learned new tools and continuously improved our skills through deliberate reflection along the way.

3.6 A Humbling Reminder: The Story of a Single Voice Speaking An Unaddressed Truth

By the public release of the app on October 1, 2020, we could look back on 13 studies, including over 1000 participants, where over 150 usability and attitudinal issues had been revealed, prioritized, and addressed as much as time and resources allowed. We were exhausted, but satisfied with our process and the results.

Upon release, we started getting feedback in the app stores and through an in-app feedback mechanism. Much of this feedback either validated the design decisions we had made or echoed issues we had discussed but did not have time to fix. We were celebrating a job well done.

One voice in the in-app feedback brought us back to earth:

“As a nurse working in the hospital, I’m concerned on how to differentiate false exposures (in contact with a COVID+ patient while wearing correct PPE) with true exposures (in contact with a COVID+ patient while not wearing the correct PPE).”

This was a humbling reminder to not get so enamored of our data that we forget its limitations. Our process had missed important needs of front-line healthcare providers. During the preparations for each empirical study, we attempted to include a diversity of participants, but were not often successful. The UserZoom panel allowed us to recruit beyond “friends & family,” expanding to some aspects of diversity, such as different age groups and geographic localities, but not for others, like profession. Although we made several attempts, we were not able to establish the trusted relationships required to work with important user groups, like hospital personnel and communities at risk, within the tight deadlines of this project.

At this writing, we are closely monitoring the real-world feedback (e.g., at least four similar requests have come in from healthcare workers) and are brainstorming design responses to fill gaps we did not anticipate.

4 CONCLUSIONS

The project to design and evaluate *COVID Alert NY* provided many lessons for UX practice, six of which appear here. Consider starting projects with an analytic method to get many benefits fast. Make do with the situation you have, as UX methods can often be modified. Not all methods stretch to all situations, so expect some failures and be prepared to move on. Sometimes a small, quick design change can have huge benefits. Improve your skills as you go. Finally, remember to be humble about your data; the real world will always win.

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